



Lesson 2.6 — The Quadratic Formula

Big idea: The quadratic formula solves any quadratic $ax^2 + bx + c = 0$ in one step. The discriminant tells you, before you compute, how many solutions to expect.

Quick review

The formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Discriminant $D = b^2 - 4ac$:

- $D > 0 \rightarrow$ two real solutions
- $D = 0 \rightarrow$ one (repeated) real solution
- $D < 0 \rightarrow$ two complex (non-real) solutions

Use it when: factoring isn't obvious, or when you need exact (irrational/complex) answers.

Always: set the equation equal to zero before identifying a , b , c .

Worked example

Worked example. Solve $2x^2 - 4x - 3 = 0$.

$a = 2$, $b = -4$, $c = -3$. $D = 16 + 24 = 40$. $x = \frac{4 \pm \sqrt{40}}{4} = \frac{4 \pm 2\sqrt{10}}{4} = \frac{2 \pm \sqrt{10}}{2}$.

Warm-ups

1. State a , b , c for $3x^2 - 7x + 2 = 0$.
2. Compute the discriminant of $x^2 + 5x + 6 = 0$ and predict the number of real solutions.
3. Solve $x^2 - 4x - 5 = 0$ with the quadratic formula.

Practice

1. Solve $2x^2 + 3x - 1 = 0$ with the quadratic formula. Leave answer exact.
2. How many real solutions does $4x^2 - 12x + 9 = 0$ have? Find them.
3. Solve $x^2 + 4x + 13 = 0$. (Use complex numbers.)

Challenge

1. For what value(s) of k does $kx^2 + 6x + k = 0$ have exactly one real solution?
2. A ball is thrown upward from a 50 ft cliff with initial velocity 32 ft/s. Its height (ft) after t seconds is $h(t) = -16t^2 + 32t + 50$. When does it hit the ground?



Answer key — Lesson 2.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $a = 3$, $b = -7$, $c = 2$.
2. $D = 25 - 24 = 1 > 0 \rightarrow$ two real solutions.
3. $x = (4 \pm \sqrt{36})/2 = (4 \pm 6)/2 = 5$ or -1 .

Practice

1. $x = (-3 \pm \sqrt{17})/4$.
2. $D = 144 - 144 = 0 \rightarrow$ one repeated solution. $x = 12/8 = 3/2$.
3. $D = 16 - 52 = -36$. $x = (-4 \pm 6i)/2 = -2 \pm 3i$.

Challenge

1. $D = 36 - 4k^2 = 0 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$. (Reject $k = 0$ because then it isn't quadratic.)
2. Set $h = 0$: $-16t^2 + 32t + 50 = 0 \Rightarrow 16t^2 - 32t - 50 = 0 \Rightarrow 8t^2 - 16t - 25 = 0$. $t = (16 \pm \sqrt{(256 + 800)})/16 = (16 \pm \sqrt{1056})/16 \approx (16 \pm 32.50)/16$. Take positive: $t \approx 3.03$ seconds.